

Brandon Knutson

PhD Candidate — ML & Applied Mathematics

☎ (720) 499-9097 ✉ bknutson314@gmail.com 🔗 brandon-knutson-8102 🔄 bknutson0 🌐 bknutson0.github.io

EDUCATION

Colorado School of Mines <i>PhD Computational & Applied Mathematics</i> Advisors: Dr. Daniel McKenzie & Dr. Samy Wu Fung	Golden, CO 2022–Present
University of Colorado Boulder <i>BA Mathematics & Physics (with Distinction)</i>	Boulder, CO 2014–2018

EXPERIENCE

Oak Ridge National Laboratory <i>Machine Learning Intern</i> ○ Trained XGBoost model on 898k buildings to predict heights from satellite imagery (0.70m MAE). Built data-processing pipeline for vector/raster geospatial data using PostGIS, QGIS, and Docker for scalable height estimation across urban, suburban, and rural environments.	Oak Ridge, TN 2023
Colorado School of Mines <i>Instructor & Teaching Assistant (Differential Equations)</i> ○ Instructor of Record (2024) with course design responsibilities, after leading recitations and office hours as TA (2022–2023).	Golden, CO 2022–Present
US Navy <i>Operations Research Analyst (GS-11)</i> ○ Developed data-intensive cost models to evaluate acquisition proposals up to \$4.1B. Communicated mathematical analyses to senior decision-makers, securing hundreds of millions in savings. Secret clearance.	Norfolk, VA 2019–2022

RESEARCH

- **Faster Implicit Network Training via Random Truncation** *Manuscript in preparation*
Theoretical and empirical analysis of a new class of training methods for implicit networks. Random truncation reduces computational cost while preserving convergence.
- **Logical Extrapolation in Neural Networks** *AAAI 2026 (Oral presentation)*
First-author paper at top-tier AI conference benchmarking how implicit and recurrent neural networks generalize to harder, out-of-distribution mazes. Error analysis revealed models learning algorithmic shortcuts that failed on harder inputs. Slight training-data diversification dramatically improved accuracy. [On Logical Extrapolation for Mazes with Recurrent and Implicit Networks](#).
- **Open-Source Maze Generation Package** *Journal of Open Source Software*
Co-developed peer-reviewed Python package with 80+ GitHub stars, used by research community to generate datasets for studying neural network reasoning. [maze-dataset: Maze Generation with Algorithmic Variety and Representational Flexibility](#).

SKILLS

Programming & Tools: Python, SQL, PyTorch, Pandas, Scikit-learn, MATLAB, Mathematica, $\text{\LaTeX}$, Jupyter Notebooks, Git, Docker, AWS, Google Cloud, Linux, HPC, Multi-GPU Systems, AI Coding Assistants (Claude Code, GitHub Copilot)

Core Competencies: Deep Learning, Implicit & Recurrent Neural Networks, Bi-level Optimization, Stochastic Methods, Linear Algebra, Topological Data Analysis, Data Visualization

AWARDS & HONORS

2014–2018: GPA: 3.99, Dean's List, Phi Beta Kappa, Pi Mu Epsilon

2018: Marlene Massaro Pratto and David Pratto Scholarship (Mathematics)